

Threat Model (Password Risk Intelligence System)

Parameters

- Password input by the User.
 - Derived Risk Analytics.
 - Database logs.
 - API communication (HaveIbeenPwned).
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Threats

1. Password Leakage

Mitigation:

- User Passwords are not stored in the Database.
 - Password is Hashed and then sent through API requests.
 - Only derived metrics are logged and Visualized.
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2. API

Mitigation:

- HTTPS Protocol used.
 - Only Prefix Exposure after Hashing.
 - Concept of K-anonymity implemented.
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3. SQL Injection

Mitigation:

- Table Queries are Parameterized.
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4. Denial of Service (DDoS)

Mitigation:

- API timeout after each request sent.
 - Exception handling regarding API requests.
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5. Dashboard Data Exposure

Mitigation:

- Deployment of Dashboard locally.
- No sensitive data (User Passwords) is stored.